

Business Problem Statement

“Short-form videos dominate digital consumption, yet only a small percentage go viral. Brands and creators lack clear, data-backed guidelines on what actually increases views, retention, and early engagement.

This project analyzes reel/short-form video performance data to **identify the key factors** that drive **virality** and **provide actionable recommendations for maximizing reach and audience retention.**”

Deliverables (End-to-End Project Workflow)

1. Data Import & Initial Understanding

- Import the dataset into Python to explore its structure, data types, and completeness.
- Perform an initial review to understand the variables related to video performance.

2. Data Cleaning & Preparation (Python)

- Clean and standardize the dataset by fixing data types, handling missing values, and formatting date/time fields.
- Create additional relevant columns if needed.
- Remove redundant or irrelevant columns to make the dataset analysis-ready.

3. Data Transformation & Organization

- Organize the cleaned dataset into a structured format suitable for SQL analysis.
- Export or load the transformed dataset into MySQL for deeper analytical work.

4. Exploratory Analysis Using SQL

- Use MySQL to run structured, query-based analysis to uncover patterns, trends, and performance indicators.
- Perform filtering, grouping, aggregation, and ranking queries to extract meaningful insights.
- Validate patterns identified during Python exploration through SQL-based breakdowns.

5. Visualization & Dashboard Development (Power BI)

- Connect the SQL database to Power BI to build an interactive dashboard.

- Create visualizations that highlight key metrics such as retention, views, niche performance, music type impact, and upload timing.
- Add filters and slicers to allow interactive exploration of insights.

6. Analytical Report & Insights Summary

- Compile all findings into a well-structured report summarizing insights, trends, and recommendations.
- Use visuals, tables, and narrative explanation to present conclusions clearly and professionally.
- Provide actionable insights based on the data for improving reel/short content performance.

7. Final Deliverables Packaging

- Consolidate the Python scripts, SQL files, dashboard, and report into a cohesive project folder for portfolio use.
- Prepare a summary-ready version for GitHub/LinkedIn showcasing the workflow and outcomes.